

Final Project: Leading Factors for R&D Investments

DATA 201 - Spring 2026

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Motivation

A country investment in R&D depends on various factors: wealth, technological and human development, policy, uncertainty, etc. In project 2, we concluded that the dominant factor was wealth (GDP/capita). However, it didn't explain the **inconsistency** between ranking of countries by GDP/capita and their real R&D dominance in the world.

We decided to include 2 additional factors: researchers/per-million and govt_effectiveness.

QUESTION:

Which are the most relevant factors determining R&D spending and global dominance?

Datasets

World Bank - Development Indicators (WBI) - 6 indicators

<https://databank.worldbank.org/source/world-development-indicators>

World Bank - Worldwide Governance Indicators - gov_effectiveness

<https://www.worldbank.org/en/publication/worldwide-governance-indicators>

(tracks quality of public services, the civil service, policy formulation and implementation, and the credibility of governments decisions)

World uncertainty index (WUI)

<http://worlduncertaintyindex.com/data/>

(tracks uncertainty (sentiment analysis) quarterly across the globe since 1950s. 143 countries.)

(8 indicators - timeperiod: 1995-2024, 52 selected countries.)

Bias: sparse information for the less developed or more restrictive countries. The 52 countries do not include any of the low-income nations.

Merged working dataset after cleaning:

1560 rows and 13 variables

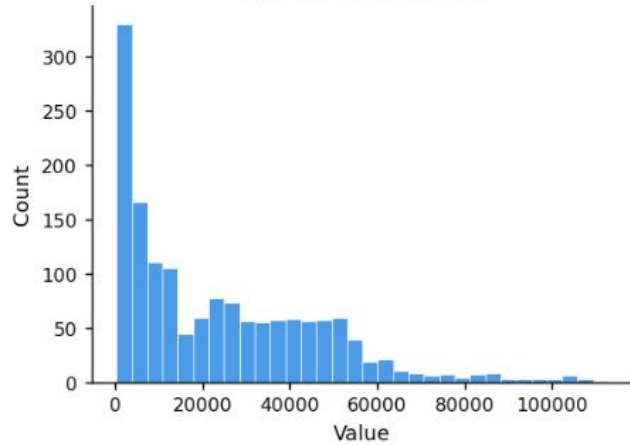
Variables of interest

Variable		Type
Researchers_per_million	(new variable)	Quantitative
Government_effectiveness	(new variable)	Quantitative
iso_code	ISO 3-letter country code	Categorical
region	(6 world regions)	Categorical
income_group	lower-middle, upper-middle, high	Categorical
GDP_per_capita	US dollars	Quantitative
Health_spend	%GDP	Quantitative
Education_spend	%GDP	Quantitative
RD_spend	%GDP	Quantitative
Life_expectancy	years	Quantitative
WUI	uncertainty index	Quantitative

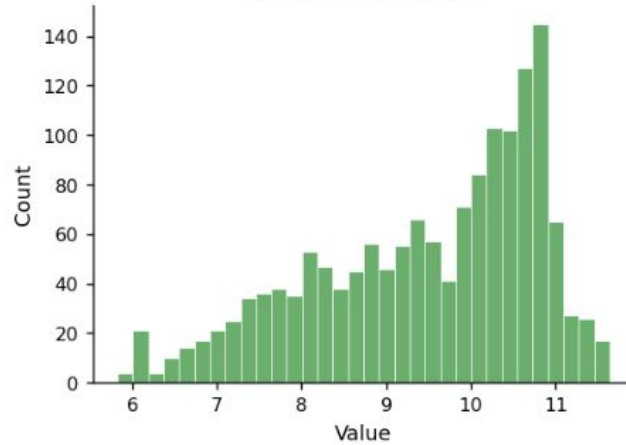
Background from Project 2

Exploratory data analysis and visualization

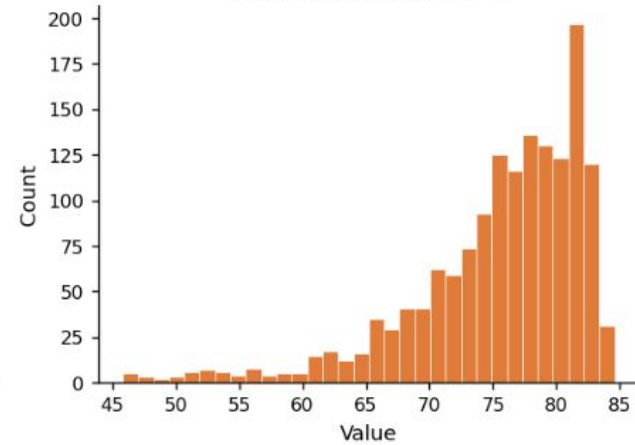
GDP per Capita (USD)



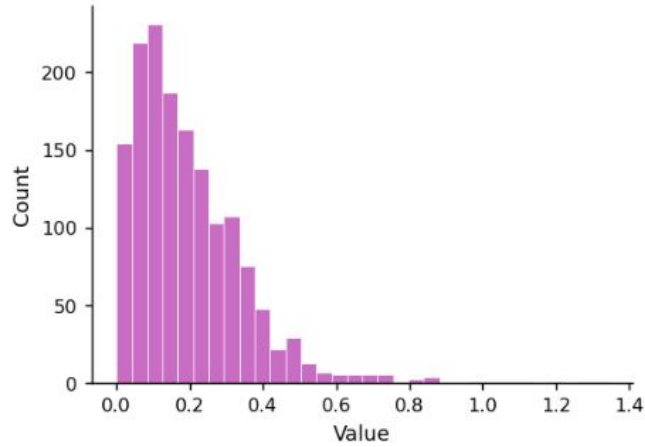
Log(GDP per Capita)



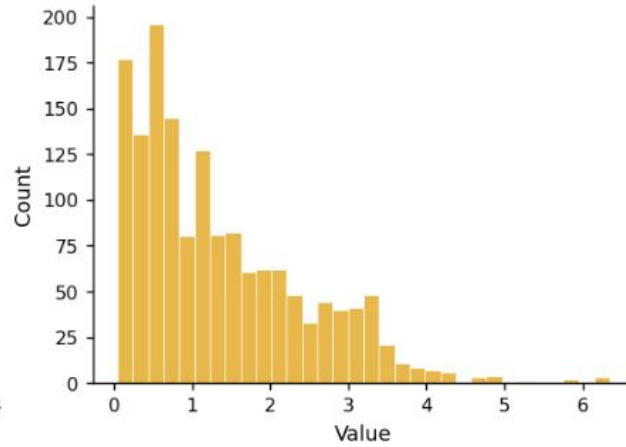
Life Expectancy (years)



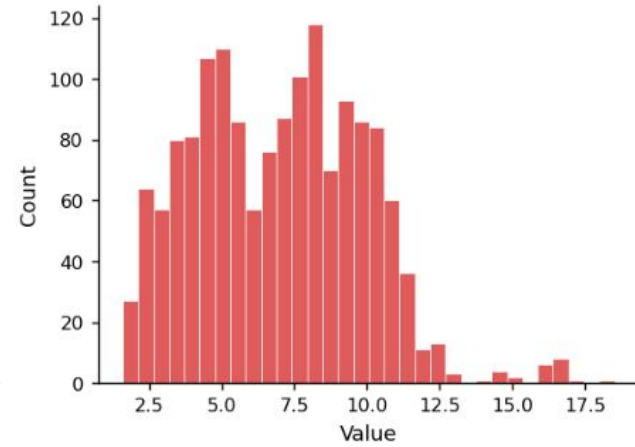
World Uncertainty Index



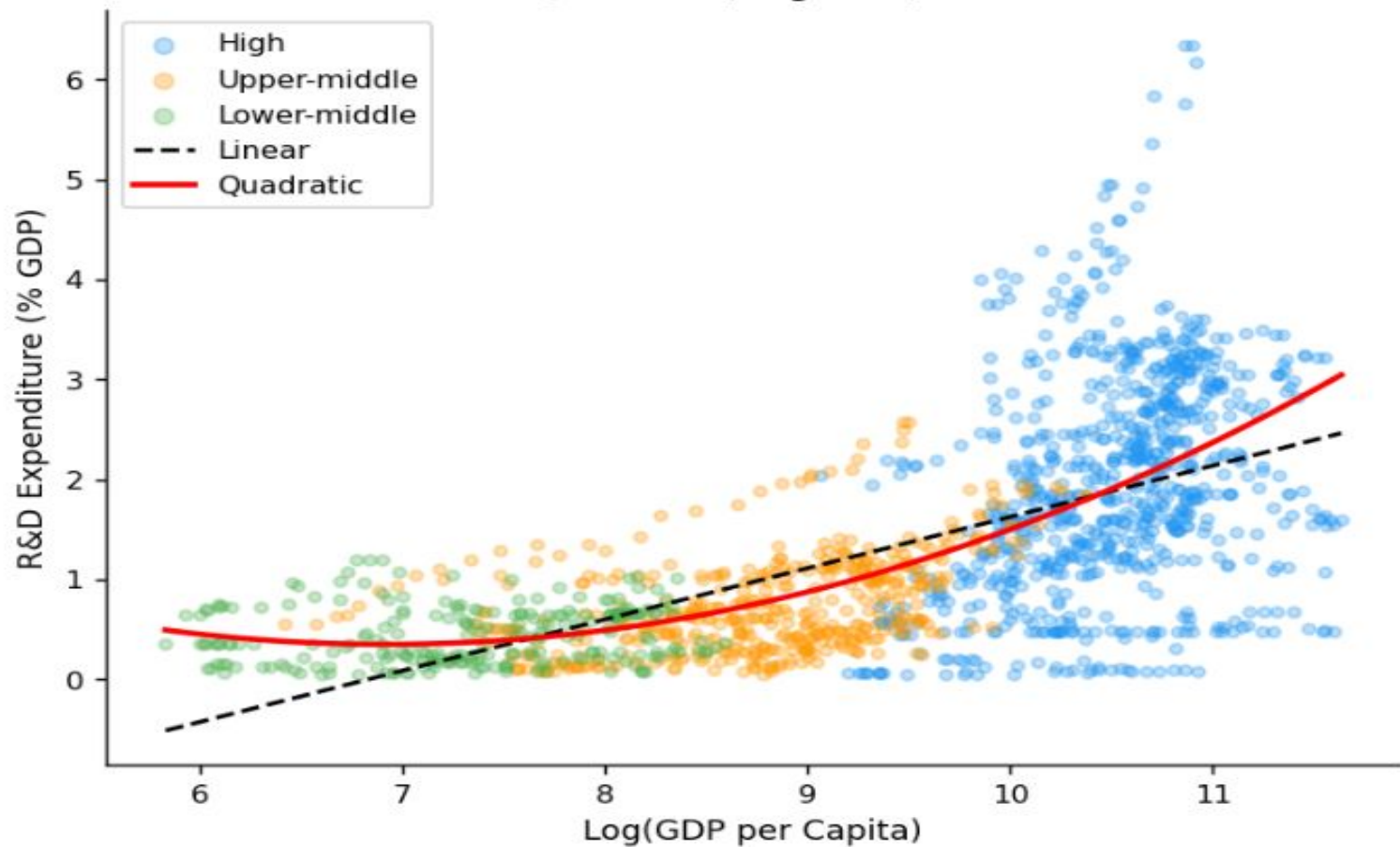
R&D Expenditure (% GDP)



Health Expenditure (% GDP)



Quadratic (Degree-2) Fit



Project 2 Model Results for prediction of R&D

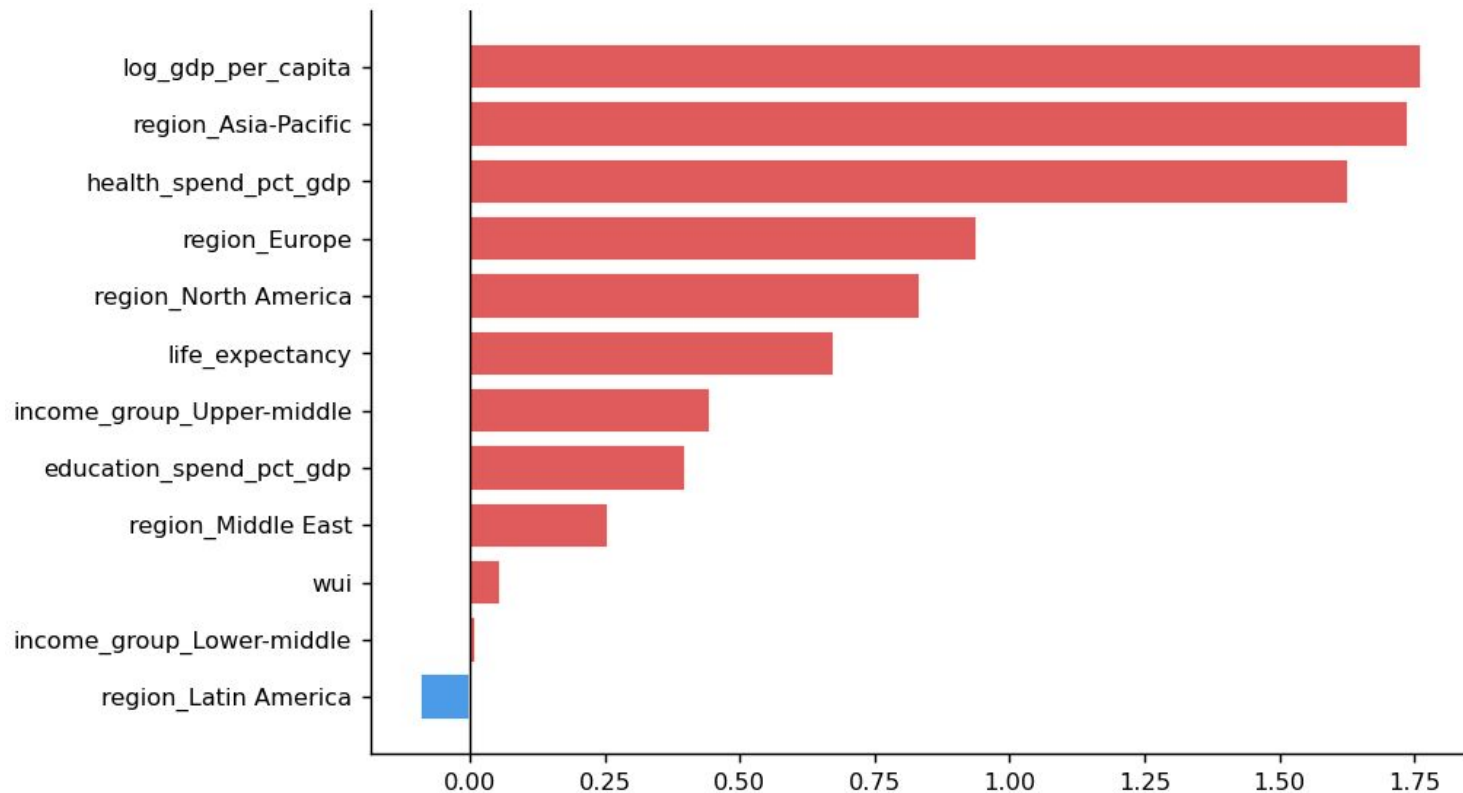
Tried 4 models:

0. Linear Regression without region and income group included
1. Linear regression including all features, including region and income group
2. Polynomial regression: quadratic in log GDP
3. Interaction of log GDP and WUI

Model Comparison:

	Model	RMSE	R2
0	Linear	0.780412	0.506989
1	Linear with Categorical	0.688461	0.616321
2	Polynomial	0.689879	0.614739
3	Interaction	0.688766	0.615981

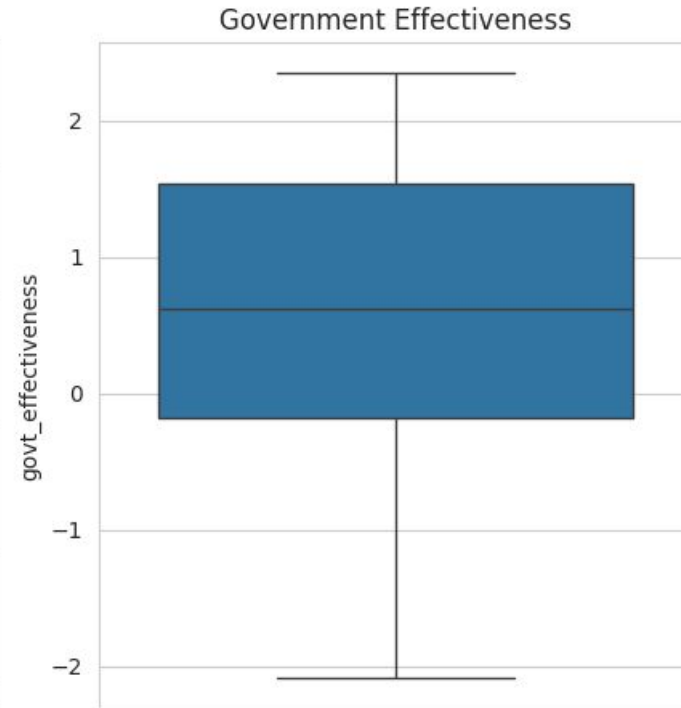
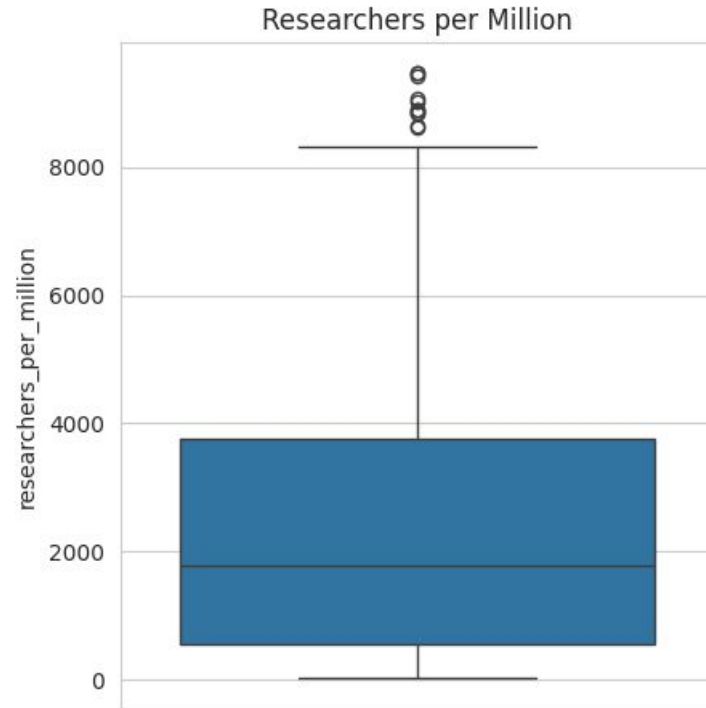
Logistic correlation coefficients vs log odds: variable importance



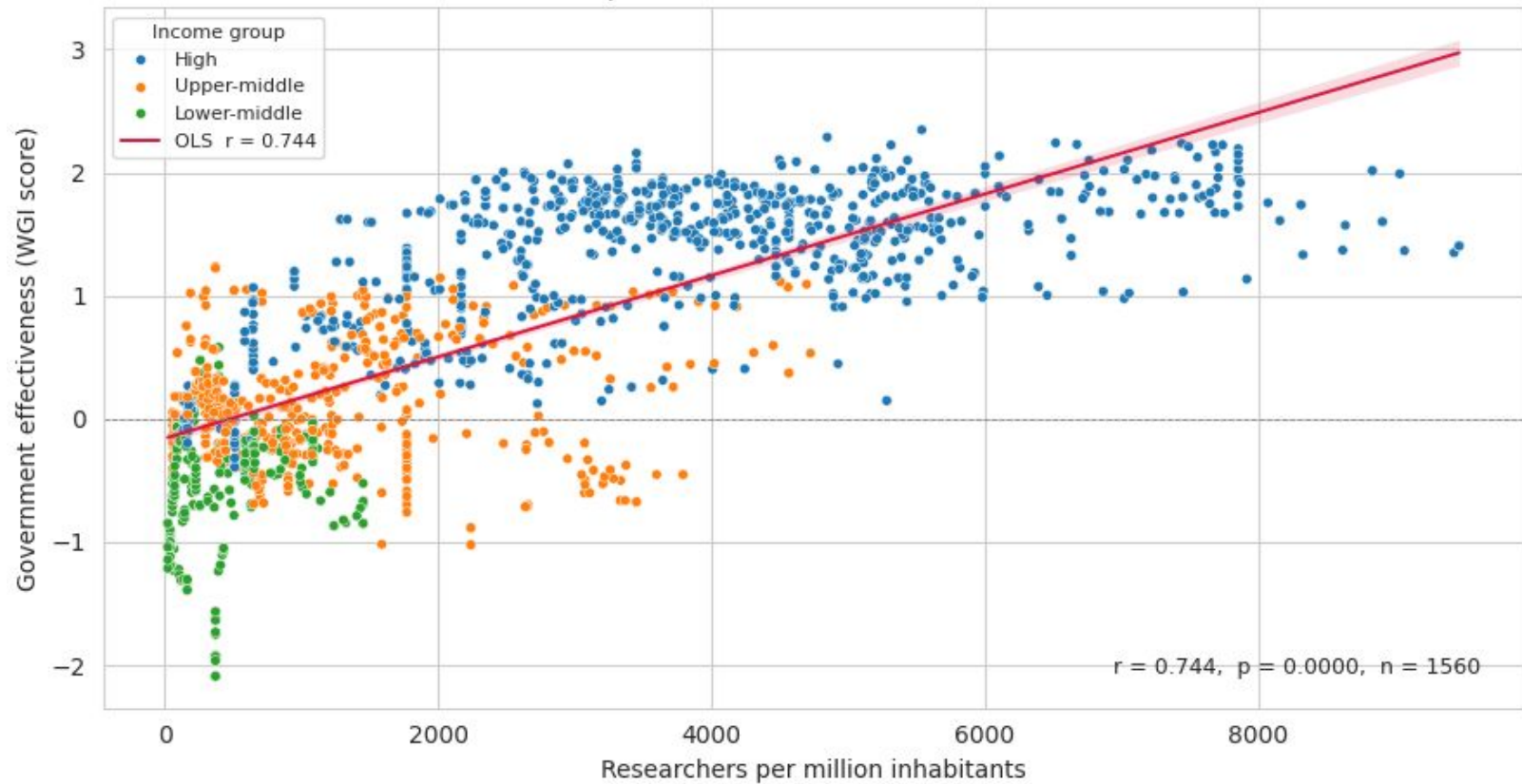
Added to Final Project

Researchers per Million	(human capital)
Government Effectiveness	(adherence to policies robust legal system)

Distribution of New Predictor Variables

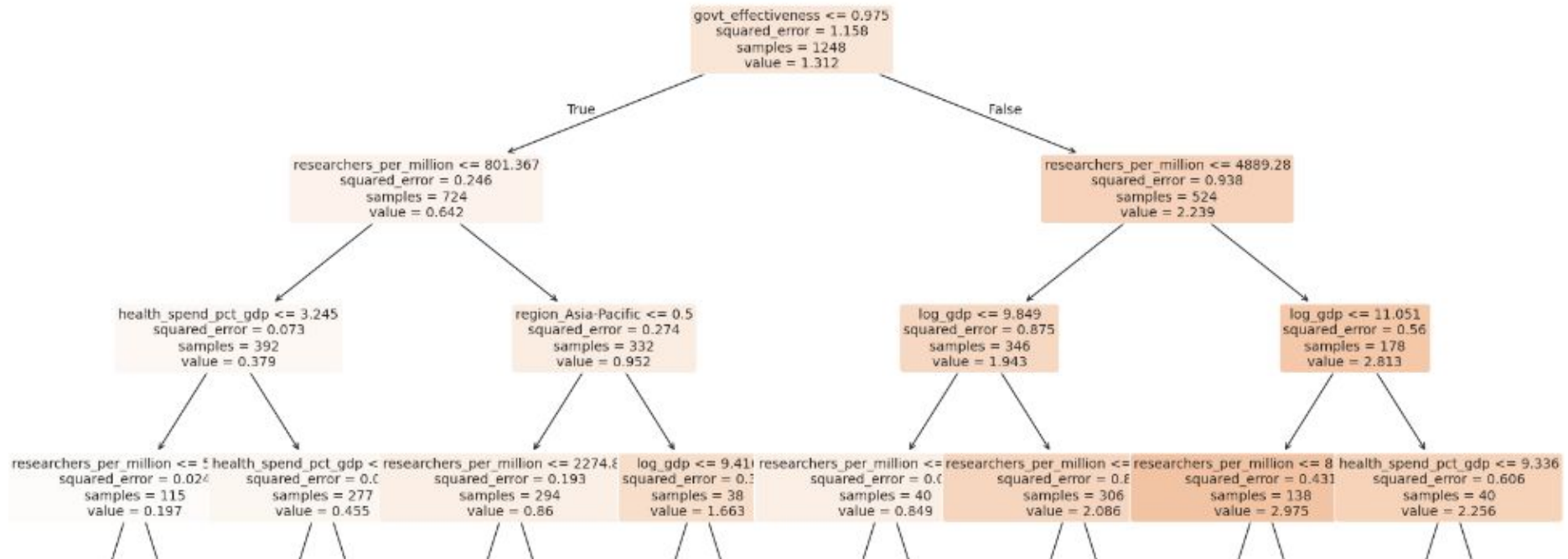


Researchers per Million vs Government Effectiveness

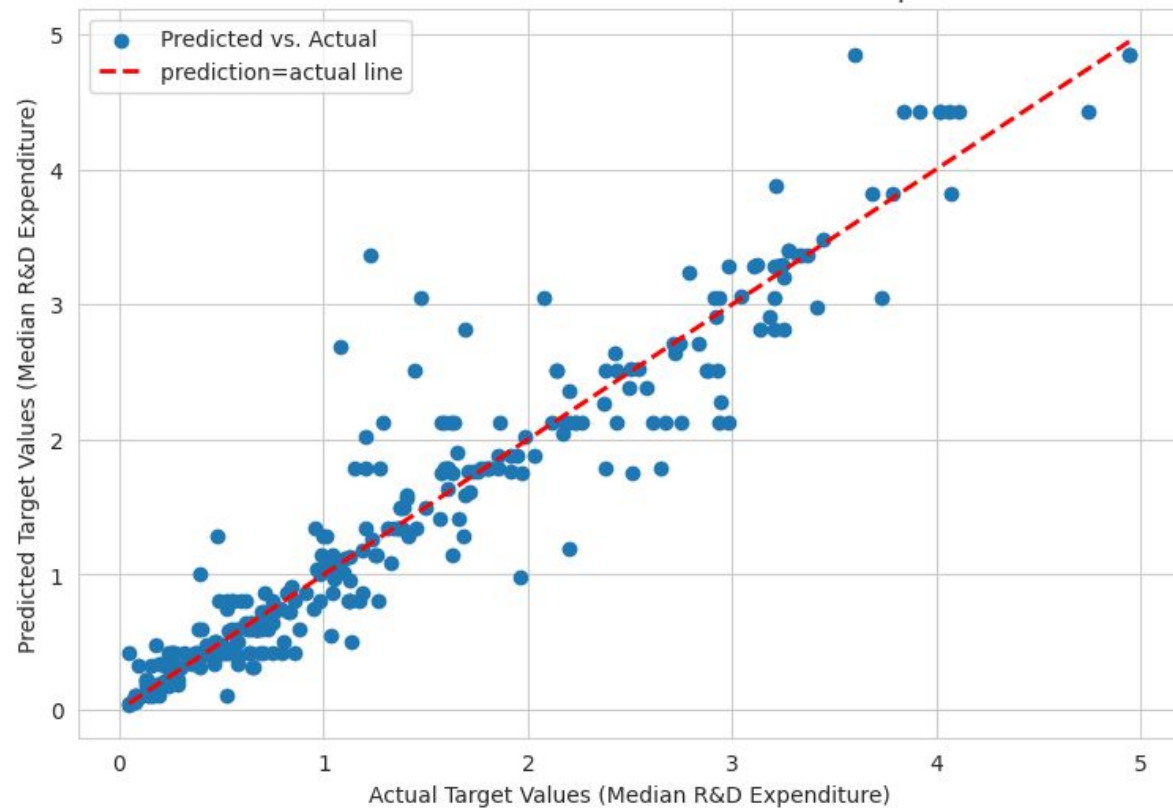


Predict R&D expenditure as %GDP using Tree Models

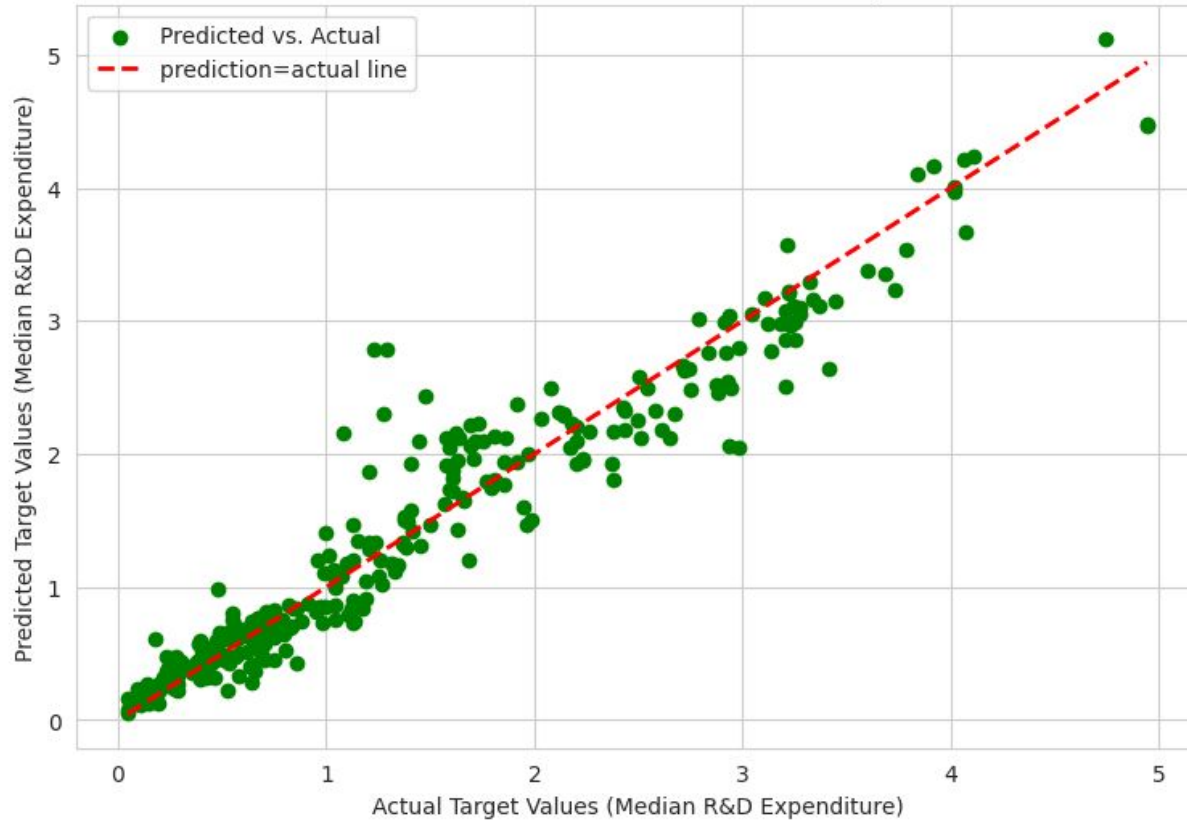
Decision Tree Structure (Visualized Depth = 3)



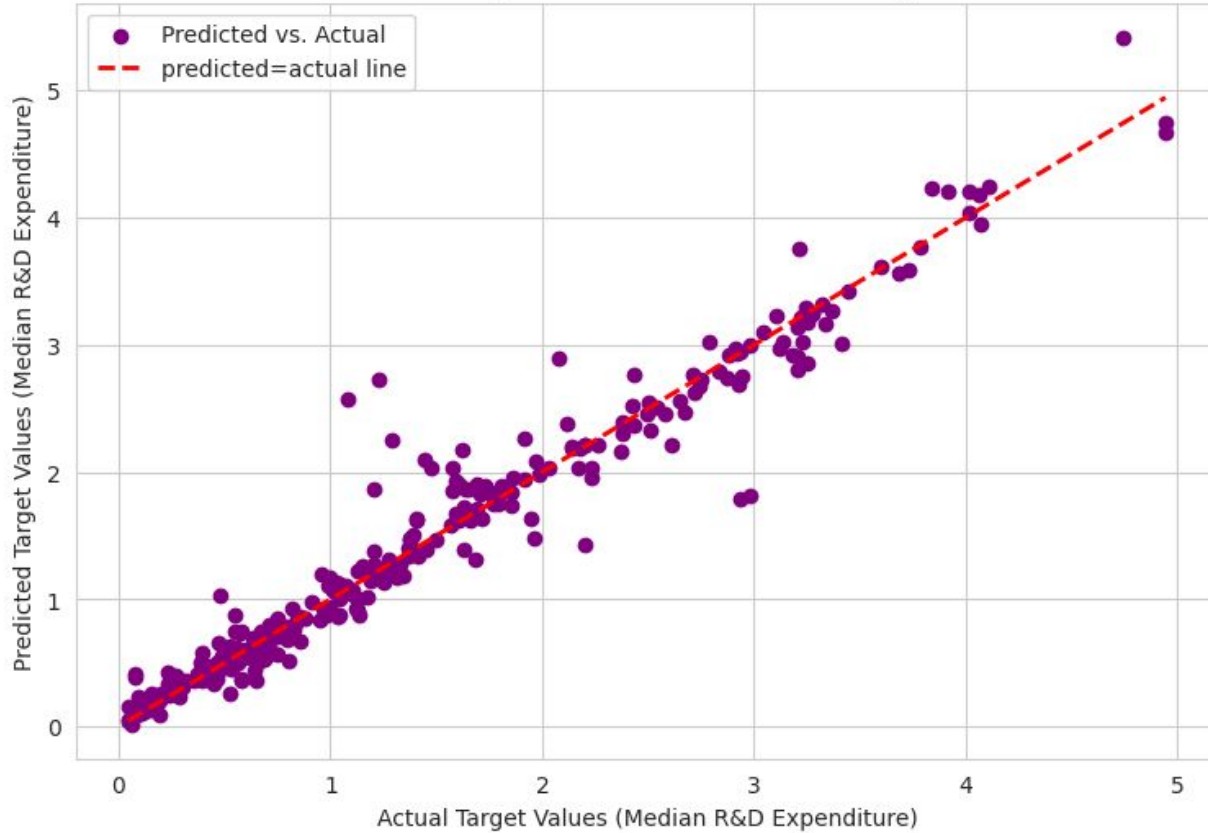
8-Level Decision Tree: Actual vs. Predicted R&D Expenditure



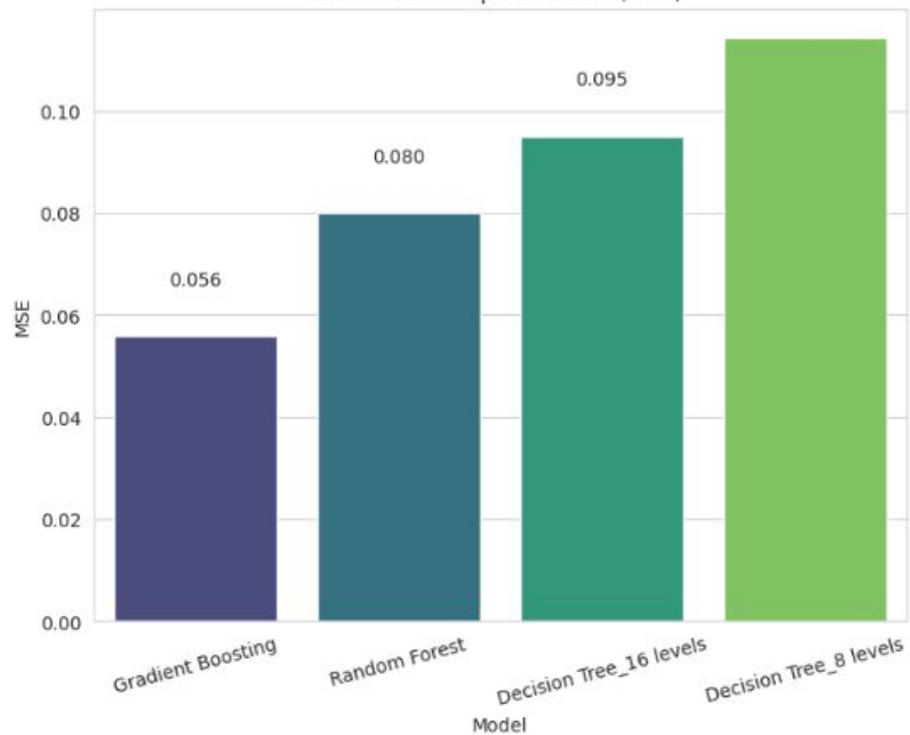
Random Forest: Actual vs. Predicted R&D Expenditure



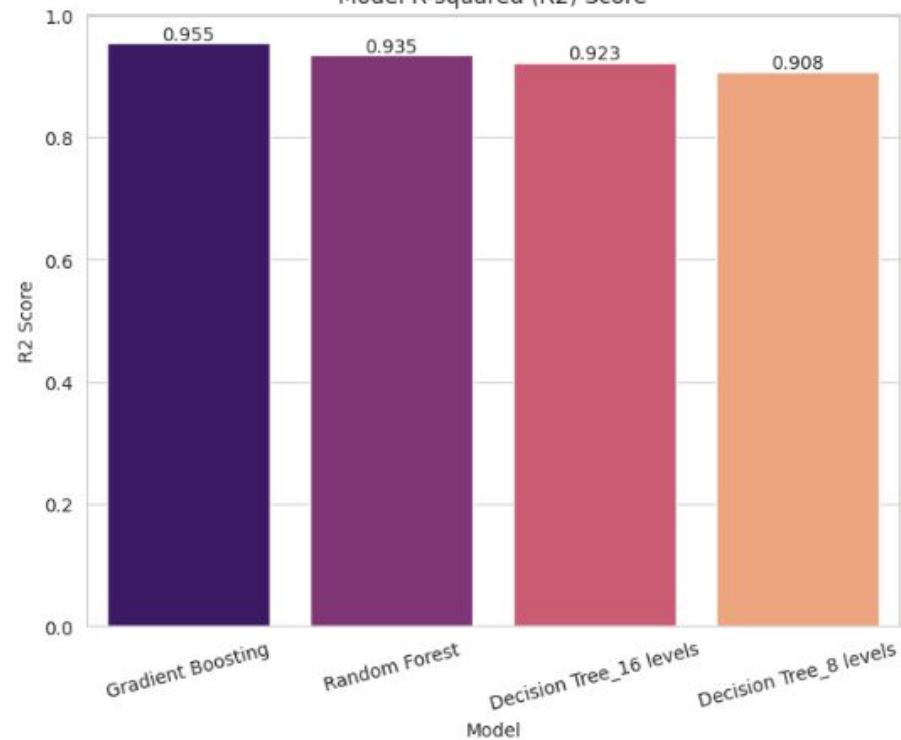
Gradient Boosting: Actual vs. Predicted R&D Expenditure



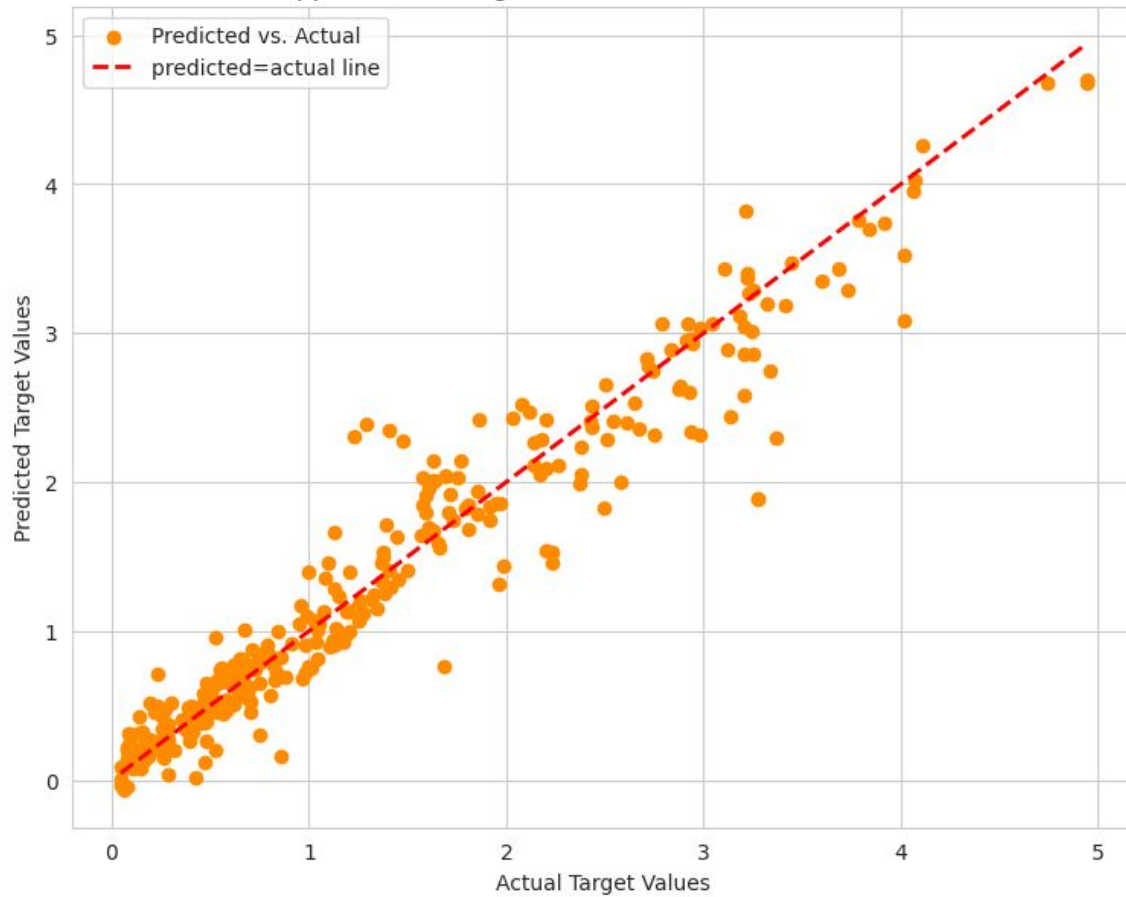
Model Mean Squared Error (MSE)



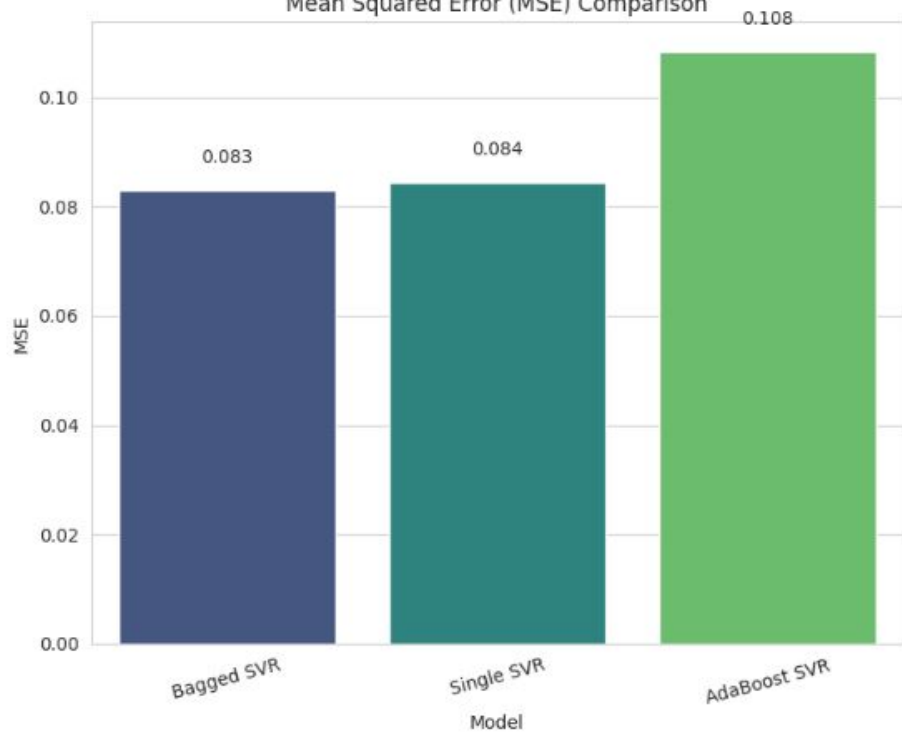
Model R-squared (R2) Score



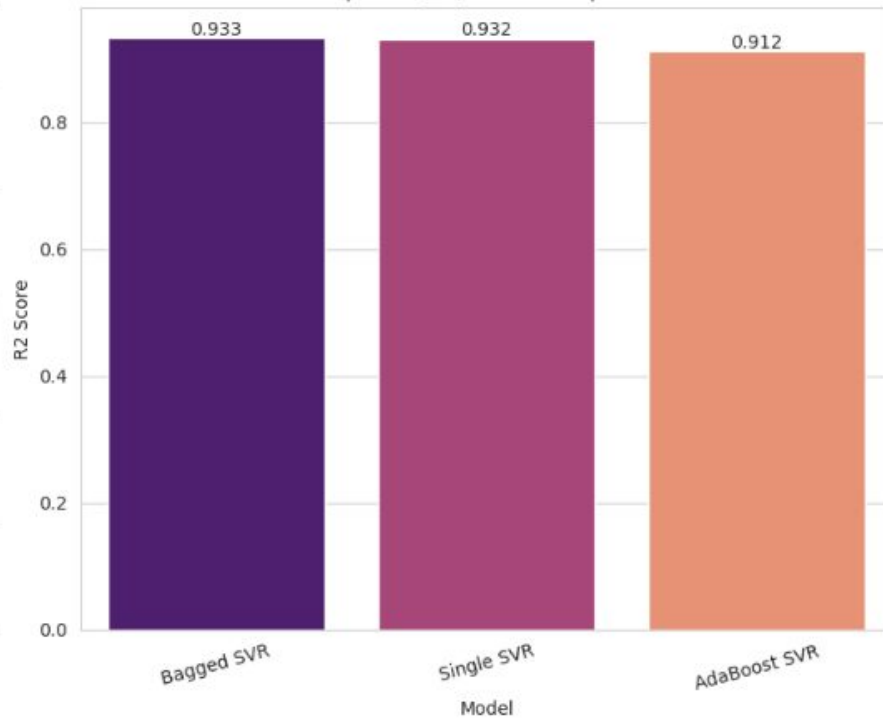
Support Vector Regressor: Actual vs. Predicted Values



Mean Squared Error (MSE) Comparison



R-squared (R^2) Score Comparison



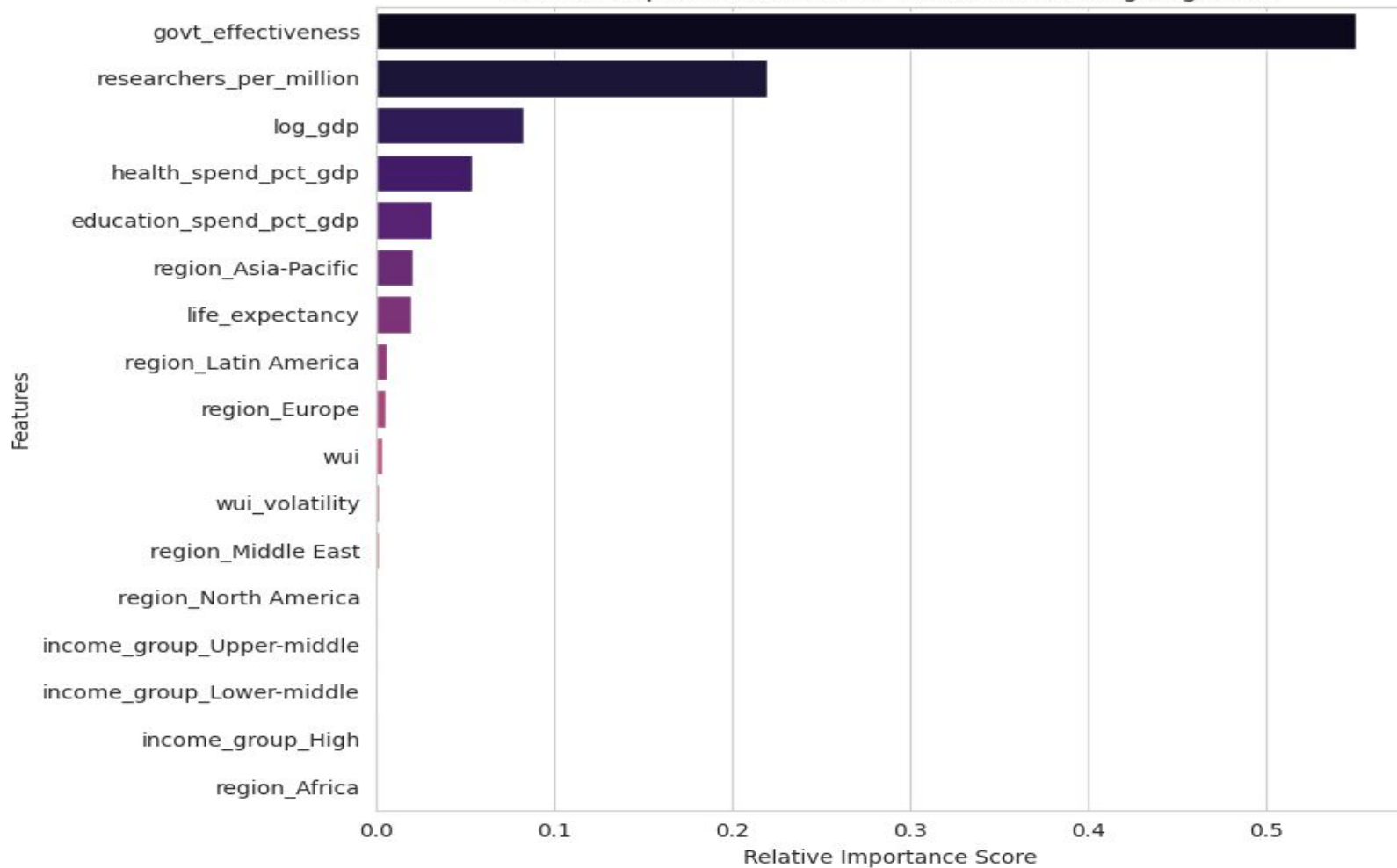
--- Model Comparison Table ---

MSE	R2	Model
0.0558855	0.954762	Gradient Boosting
0.0800327	0.935215	Random Forest
0.0950354	0.92307	Decision Tree_16 levels
0.114267	0.907503	Decision Tree_8 levels

--- Consolidated SVR Model Comparison Table ---

Model	MSE	R2
Bagged SVR	0.0830	0.9328
Single SVR	0.0845	0.9316
AdaBoost SVR	0.1083	0.9123

Feature Importance (MDI) for Gradient Boosting Regressor



Conclusions

A model to predict R&D investment as %GDP was created including as features:

Government effectiveness, country wealth (GDP/capita), human capital (researchers/million), social investment (life expectancy, health, education), regional and income differences, periodic upsets (uncertainty).

The leading factor is Government effectiveness, 2x as important as human capital, and 4x as important as GDP/capita.

The results from decision tree-based models and from support vector machine models are very robust, with $R^2 > 0.9$ in all cases.

For my dataset, DT models perform better than SVM models under comparable conditions (hyperparameter values). Gradient boosting DT yields the highest R^2 and lowest MSE. We did not observe overfitting after increasing the complexity (levels) of the DT model.